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MS5032 Financial Markets and Derivatives - Quiz 1 - February 2026
Total Time: 90 minutes; Total Marks :50
Section A (30*1=30 Marks; Maximum Time 30 minutes)

22/30

1. Which of the following is **not** a characteristic of a money market instrument?

A. Liquidity
B. Marketability
C. Long maturity
D. Liquidity premium
E. Long maturity and liquidity premium

2. Commercial paper is a short-term security issued by ________ to raise funds.

A. the Federal Reserve Bank
B. commercial banks
C. large, well-known companies
D. the New York Stock Exchange
E. state and local governments

3. Which of the following statements is **true** regarding a corporate bond?

A. A corporate callable bond gives the holder the right to exchange it for a specified number of the company's common shares.
B. A corporate debenture is a secured bond.
C. A corporate indenture is a secured bond.
D. A corporate convertible bond gives the holder the right to exchange the bond for a specified number of the company's common shares.
E. Holders of corporate bonds have voting rights in the company.

4. Which of the following is **true** regarding a firm's securities?

A. Common dividends are paid before preferred dividends.
B. Preferred stockholders have voting rights.
C. Preferred dividends are usually cumulative.
D. Preferred dividends are contractual obligations.
E. Common dividends can usually be paid if preferred dividends have been skipped.

5. The money market is a subsector of the

A. commodity market.
B. capital market.
C. derivatives market.
D. equity market.
E. None of the options are correct.

$(\frac{1}{4})$ $\frac{1}{32}$ $\frac{1}{8}$

6. Which one of the following is **not** a money market instrument?

A. Treasury bill
B. Negotiable certificate of deposit
C. Commercial paper
D. Treasury bond
E. Eurodollar account

7. The price quotations of Treasury bonds in the *Wall Street Journal* show an ask price of 104.25 and a bid price of 104.125. As a buyer of the bond, what is the dollar price you expect to pay?

A. $1,048.00
B. $1,042.50
C. $1,044.00
D. $1,041.25
E. $1,040.40

8. You purchased JNJ stock at $50 per share. The stock is currently selling at $65. Your gains may be protected by placing a

A. stop-buy order.
B. limit-buy order.
C. market order.
D. limit-sell order.
E. None of these options are correct.

9. Assume you purchased 200 shares of GE common stock on margin at $70 per share from your broker. If the initial margin is 55%, how much did you borrow from the broker?

I
14 x 0.45
630

A. $6,000
B. $4,000
C. $7,700
D. $7,000
E. $6,300

10. Which of the following orders is most useful to short sellers who want to limit their potential losses?

A. Limit order
B. Discretionary order
C. Limit-loss order
D. Stop-buy order

11. Which of the following is a criterion for admission of a stock into S&P CNX Nifty

a. Stock with low P/E
b. Stocks with low impact cost
c. Stocks where rating agencies have up graded
d. Volume of trade

12. Investment bankers

A. act as intermediaries between issuers of stocks and investors.
B. act as advisors to companies in helping them analyze their financial needs and find buyers for newly-issued securities.
C. accept deposits from savers and lend them out to companies.
D. act as intermediaries between issuers of stocks and investors and act as advisors to companies in helping them analyze their financial needs and find buyers for newly-issued securities.

13. Which one of the following statements regarding orders is **false**?

A. A market order is simply an order to buy or sell a stock immediately at the prevailing market price.

B. A limit-sell order is where investors specify prices at which they are willing to sell a security.

C. If stock ABC is selling at $50, a limit-buy order may instruct the broker to buy the stock if and when the share price falls below $45.

D. A market order is an order to buy or sell a stock on a specific exchange (market).

14. Which one of the following statements regarding open-end mutual funds is **false?**

A. The funds redeem shares at net asset value.
B. The funds offer investors professional management.
C. The funds offer investors a guaranteed rate of return.
D. The funds redeem shares at net asset value and offer investors professional management.

15. Which of the following statements about money market mutual funds is **true**?

A. They invest in commercial paper, CDs, and repurchase agreements.
B. They usually offer check-writing privileges.
C. They are highly leveraged and risky.
D. They invest in commercial paper, CDs, and repurchase agreements, and they usually offer check-writing privileges.
E. All of the options are true.

16. Funds which invest only in the stocks comprising an index and aim to give returns commensurate with the index returns are called ------------

a) Equity Funds
b) Exchange traded funds
c) Money market funds
d) Index Funds

17. Closed-end funds are frequently issued at a ______ to NAV and subsequently trade at a __________ to NAV.

A. discount; discount
B. discount; premium
C. premium; premium
D. premium; discount
E. No consistent relationship has been observed.

18. Diversified Portfolios had year-end assets of $279,000,000 and liabilities of $43,000,000. If Diversified's NAV was $42.13, how many shares must have been held in the fund?

A. 43,000,000
B. 6,488,372
C. 5,601,709
D. 1,182,203

19 .________ funds do not have a fixed date of redemption.

a) Open ended funds
b) Close ended funds
c) Diversified funds
d) Equity Funds

20. The current yield on a bond is equal to

A. annual interest payment divided by the current market price.
B. the yield to maturity.
C. annual interest divided by the par value.
D. the internal rate of return.
E. None of the options are correct.

21. If a 7% coupon bond is trading for \$975.00, it has a current yield of

A. 7.00%.
B. 6.53%.
C. 7.24%.
D. 8.53%.
E. 7.18%.

22. A firm with a low rating from the bond-rating agencies would have

A. a low times-interest-earned ratio.
B. a low debt-to-equity ratio.
C. a low quick ratio.
D. a low debt-to-equity ratio and a low quick ratio.
E. a low times-interest-earned ratio and a low quick ratio.

23. Structure of interest rates is

A. the relationship between the rates of interest on all securities.
B. the relationship between the interest rate on a security and its time to maturity.
C. the relationship between the yield on a bond and its default rate.
D. All of the options are correct.
E. None of the options are correct.

24. Treasury STRIPS are

A. securities issued by the Treasury with very long maturities.
B. extremely risky securities.
C. created by selling each coupon or principal payment from a whole Treasury bond as a separate cash flow.
D. created by pooling mortgage payments made to the Treasury.

25. NSE's screen-based trading system (SBTS) matches orders in the ________ priority basis.

(a) time/price
(b) price/time
(c) price/quantity
(d) quantity/price

26. The duration of a bond is a function of the bond's
A. coupon rate.
B. yield to maturity.
C. time to maturity.
D. All of the options are correct.
E. None of the options are correct.

27. Holding other factors constant, the interest-rate risk of a coupon bond is higher when the bond's

A. term to maturity is lower.
B. coupon rate is higher.
C. yield to maturity is lower.
D. current yield is higher.
E. None of the options are correct.

28. The interest-rate risk of a bond is

A. the risk related to the possibility of bankruptcy of the bond's issuer.
B. the risk that arises from the uncertainty of the bond's return caused by changes in interest rates.
C. the unsystematic risk caused by factors unique in the bond.
D the risk related to the possibility of bankruptcy of the bond's issuer, and the risk that arises from the uncertainty of . the bond's return caused by changes in interest rates.
E. All of the options are correct.

29. You wish to earn a return of 11% on each of two stocks, C and D. Stock C is expected to pay a dividend of $3 in the upcoming year while stock D is expected to pay a dividend of $4 in the upcoming year. The expected growth rate of dividends for both stocks is 7%. The intrinsic value of stock C

A. will be greater than the intrinsic value of stock D.
B. will be the same as the intrinsic value of stock D.
C. will be less than the intrinsic value of stock D.
D. will be the same or greater than the intrinsic value of stock D.
E. None of the options.

30. Low Tech Company has an expected *ROE* of 10%. The dividend growth rate will be ________ if the firm follows a policy of paying 40% of earnings in the form of dividends.

A. 6.0%
B. 4.8%
C. 7.2%
D. 3.0%

M5032 Financial Markets and Derivatives - Quiz 1 - February 2026
Total Time: 90 minutes; Total Marks :50
Section B (4*5=20 Marks; Maximum Time 60 minutes)

Question 1

20-year maturity bond with par value of $1,000 makes semi-annual coupon payments at a coupon rate of 8%. Find the bond equivalent and effective annual yield to maturity of the bond if the bond price is:

a. $950.

b. $1,000.

c. $1,050.

Question 2

The following is a list of prices for zero-coupon bonds of various maturities. Calculate the yields to maturity of each bond and the implied sequence of forward rates.

Maturity (Years)	Price of Bond ($)
1	943.40
2	898.47
3	847.62
4	792.16

Question 3

Philip Morris has issued bonds that pay semi-annually with the following characteristics

Coupon	Yield to Maturity	Maturity	Macaulay Duration
8%	8%	15 years	10 years

a. Calculate modified duration using the information above.

b. Explain why modified duration is a better measure than maturity when calculating the bond's sensitivity to changes in interest rates.

$$\frac{\Delta P}{P} = -\left(\frac{D}{1+y}\right) \cdot (1+\Delta y)$$

c. Identify the direction of change in modified duration if: i. The coupon of the bond were 4%, not 8%. ii. The maturity of the bond were 7 years, not 15 years.

d. Define convexity and explain how modified duration and convexity are used to approximate the bond's percentage change in price, given a change in interest rates

Question 4

The FI Corporation's dividends per share are expected to grow indefinitely by 5% per year.

a. If this year's year-end dividend is $8 and the market capitalization rate is 10% per year, what must the current stock price be according to the DDM?

b. If the expected earnings per share are $12, what is the implied value of the ROE on future investment opportunities?

c. How much is the market paying per share for growth opportunities (i.e., for an ROE on future investments that exceeds the market capitalization rate)?

MS5032 Financial Markets and Derivatives - Quiz 2 - March 2026
Total Time: 90 minutes; Total Marks :50

Question 1

At the end of one day a clearinghouse member is long 100 contracts, and the settlement price is \$50,000 per contract. The original margin is \$2,000 per contract. On the following day the member becomes responsible for clearing an additional 20 long contracts, entered into at a price of \$51,000 per contract. The settlement price at the end of this day is \$50,200. How much does the member have to add to its margin account with the exchange clearinghouse?

Question 2

Companies A and B face the following interest rates (adjusted for the differential impact of taxes)

	A	B
U.S. Dollars (Floating rate)	LIBOR + 0.5%	LIBOR + 1.0%
Canadian dollars (Fixed rate)	5.0%	6.5%

Assume that A wants to borrow U.S. dollars at a floating rate of interest and B wants to borrow Canadian dollars at a fixed rate of interest. A financial institution is planning to arrange a swap and requires a 50-basis-point spread. If the swap is equally attractive to A and B, what rates of interest will A and B end up paying?

Question 3

A one month European put option on a non-dividend-paying stock is currently selling for \$2.50. The stock price is \$47, the strike price is \$50, and the risk-free interest rate is 6% per annum. What opportunities are there for an arbitrageur?

Question 4

The price of a European call that expires in six months and has a strike price of \$30 is \$2. The underlying stock price is \$29, and a dividend of \$0.50 is expected in two months and again in five months. The term structure is flat, with all risk-free interest rates being 10%. What is the price of a European put option that expires in six months and has a strike price of \$30?

Question 5

A call option with a strike price of \$50 costs \$2. A put option with a strike price of \$45 costs \$3. Explain how a strangle can be created from these two options. What is the pattern of profits from the strangle?

Question 6

Suppose that put options on a stock with strike prices \$30 and \$35 cost \$4 and \$7, respectively. How can the options be used to create (a) a bull spread and (b) a bear spread? Construct a table that shows the profit and payoff for both spreads.

Question 7

Calculate the price of a three-month European put option on a non-dividend-paying stock with a strike price of \$50 when the current stock price is \$50, the risk-free interest rate is 10% per annum, and the volatility is 30% per annum.

Question 8

A stock price follows geometric Brownian motion with an expected return of 16% and a volatility of 35%. The current price is \$38.

(a) What is the probability that a European call option on the stock with an exercise price of \$40 and a maturity date in six months will be exercised?

(b) What is the probability that a European put option on the stock with the same exercise price and maturity will be exercised?

Question 9

What is the delta of a short position in 1,000 European call options on silver futures? The options mature in eight months, and the futures contract underlying the option matures in nine months. The current nine-month futures price is \$8 per ounce, the exercise price of the options is \$8, the risk-free interest rate is 12% per annum, and the volatility of silver is 18% per annum.

Question 10

A financial institution has just sold 1,000 seven-month European call options on the Japanese yen. Suppose that the spot exchange rate is 0.80 cent per yen, the exercise price is 0.81 cent per yen, the risk-free interest rate in the United States is 8% per annum, the risk-free interest rate in Japan is 5% per annum, and the volatility of the yen is 15% per annum. Calculate the delta, gamma, vega, theta, and rho of the financial institution's position. Interpret each number.